

Programmer's Model of 8086

Q. Describe the programmer's model of the 8086 microprocessor. Detail the General Purpose Registers, Segment Registers, Pointer/Index Registers, and the Flag Register.

[DEF] Definition to Remember

The Programmer's Model refers to the fourteen 16-bit internal registers accessible to an assembly programmer. They are divided into four groups: General Purpose, Segment, Pointer/Index, and the Flag Register.

1. General Purpose Registers (16-bit / 8-bit split)

Can be used as one 16-bit register or two 8-bit registers (High/Low).

* AX

2. Segment Registers (16-bit)

Hold the base addresses of 64 KB memory segments to help address 1 MB of physical memory.

* CS

3. Pointer and Index Registers (16-bit)

Hold offset addresses relative to the segment registers.

* SP

4. Flag Register (16-bit)

Contains 9 active flags (6 Status, 3 Control).

* Sta

[MUST-WRITE] Must-Write Points (for 10 marks)

1. Contains fourteen 16-bit registers.
2. General Purpose: AX (Math/IO), BX (Base address), CX (Loop counter), DX (Data/IO). Can split to 8-bit.
3. Segment Registers: CS (Code), DS (Data), SS (Stack), ES (Extra). Base addresses for 64KB segments.
4. Pointers/Index: SP & BP (Stack offsets), SI & DI (String offsets), IP (Next instruction offset).
5. Flag Register: 16-bit with 9 active flags.
6. Status Flags: Carry, Parity, Aux Carry, Zero, Sign, Overflow.

7. Control Flags: Trap, Interrupt, Direction.

[RECALL] Quick Recall

Model -> General (AX/BX/CX/DX) -> Segment (CS/DS/SS/ES) -> Pointers (SP/BP/SI/DI/IP) -> Flags (6 Status: CF/PF/AF/ZF/SF/OF, 3 Control: TF/IF/DF)